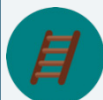
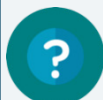


1.7 Congruence and Similarity

Lesson Introduction

 Benchmark	MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.		 Learning Targets	- I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures. - I can write congruence statements to relate corresponding parts following a rigid transformation. - I can write similarity statements to relate corresponding parts following a dilation.
	MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.			
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What determines if a shape is congruent? - What determines if a shape is similar? - How do I write a congruence statement? - How do I write a similarity statement?			
 Lesson Narrative	Students engage in identifying and describing transformations to determine which ones produce congruent figures and which ones produce similar figures. Definitions of congruent and similarity are presented, and students determine congruent and similar parts of transformed figures.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	The time needed to become an expert (<i>SE p. 41</i>)	1.7.4 Bring It Together (5 min.)	Defining congruent and similarity (<i>SE p. 45</i>)
	1.7.2 Collaboration (10 min.)	Transforming a line segment to produce a congruent or similar figure (<i>SE p. 41</i>)	1.7.5 Guided Instruction (10 min.)	Identifying transformations that produce congruent and similar images (<i>SE p. 46</i>)
	1.7.3 Guided Practice (10 min.)	Transforming angles to produce congruent angles (<i>SE p. 44</i>)	1.7.6 Wrap-Up (~5 min.)	Reflection on what was learned, what was new, and what is unclear (<i>SE p. 47</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Congruent - Similarity <u>Key Terms:</u> - N/A		- Patty Paper - Ruler (Optional) Protractor	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students do not recognize that all line segments are transformations of a preimage.	Consider if students will need to identify all congruent figures as similarities of 1:1 in your classroom. Set up the expectation early to avoid confusion and misconceptions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns Throughout the 1.7.2 Collaboration, encourage students to engage in the Take Turns routine. The partner who responds with the answer first should alternate for each question. Students must explain the rationale for their answer choice and listen to the rationale of their partner. Taking turns gives students opportunities to construct arguments and to critique the reasoning of others.	MA.K12.MTR.4.1: Discussion and Reflection In 1.7.2 Collaboration, students will work together to determine and describe transformations that produce an image from a preimage. After working together, the students share their thinking with other classmates. Students also reflect on the arguments of two sample students to determine transformations that produce congruence or similarity.
 Differentiate Instruction	Students should have an opportunity to engage with the Key Terms, similar and congruent. This will be accomplished both visually and kinesthetically if you challenge students to define each word using only pictures. Consider having students add these terms to a Word Wall. Ask students to read through these definitions and create an artistic representation of each word. The artistic approach will help students explore and retain definitions and key differences between congruence and similarity.	
Lesson Notes:		

1.7.1 Warm-Up: 10,000 Hours to Awesome

Component Overview: Students read research about the amount of time it takes for someone to become an expert at something.



1.7.1 Warm-Up: 10,000 Hours to Awesome

Best-selling author Malcolm Gladwell states in his book "Outliers" that it takes 10,000 hours of practice to be a phenom — so freakishly awesome or great at something. In 1973, Herbert Simon and William Chase wrote a paper published in *American Scientist*. Simon and Chase did experiments with chess-playing tasks and computer simulations to determine how long it would take someone to reach the grandmaster level in the game of chess. Their conclusion was that it could take as many as 50,000 hours staring at chess positions to become a chess grandmaster.

1. What skill, hobby, or activity would you like to become freakishly awesome at?

Answers will vary. I would like to become freakishly awesome at Ice Hockey.

2. How many years is 10,000 hours?

10,000 hours is just over one year, 1.14 years.

3. How many years is 50,000 hours?

50,000 hours is 5.71 years.



Consider taking the time to log student responses to question 1 so that you get to know them better.

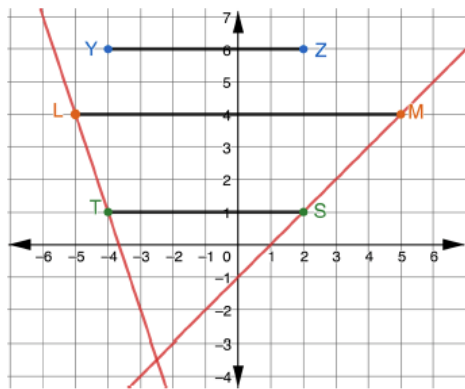
1.7.2 Collaboration: Line Segments

Component Overview: Students begin by identifying the relationship that the length of three line segments have to each other. Students then determine which transformation may have produced the different lines shown and analyze statements from others about the transformations.



1.7.2 Collaboration: Line Segments

1. $\overline{LM}$, $\overline{TS}$, and $\overline{YZ}$ are shown on the coordinate grid.



- What do you notice about the three line segments?
Answers will vary. They are all parallel. Two of the line segments are the same length.
- List at least two different ways to show that the length of $\overline{TS}$ is equal to the length of $\overline{YZ}$.
I could count the squares or measure with a ruler.
- What is the ratio of the length of $\overline{TS}$ to the length of $\overline{LM}$?
 $6:10 = 3:5$
- What is the ratio of the length of $\overline{TS}$ to the length of $\overline{YZ}$?
 $1:1$
- Describe a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{LM}$.
A single transformation to transform $\overline{TS}$ to create $\overline{LM}$ would be a dilation with a scale factor of $\frac{5}{3}$.
- Draw a line that extends through the coordinate plane and passes through points L and T .
Shown on coordinate grid.



Students may make errors during 1.7.2 Collaboration, including writing the scale factor incorrectly. Allow the opportunity for them to correct their own thinking as they work with their partner and compare their work with another partner pair in question 1j. Part of what students may realize by the end of 1.7.2 Collaboration is that the endpoints of a line segment do not have to be in the same order to consider congruence or similarity. Do not be afraid to let the mistakes happen. Wait to discuss them until the transition to the next component. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 1.7.3 Guided Practice. This could include highlighting student voices to share rich discussions or comments overhead while you circulated or to address common misconceptions that were evident among the majority of the class.



Keep in mind that some students learn ratio notation as $a:b$, a to b , or $\frac{a}{b}$.



While circulating listen for students who may incorrectly state the scale factor as $\frac{3}{5}$ instead of $\frac{5}{3}$. Address this misconception so that it is not carried forward in their learning. Consider the following questions:

- Which line segment is the preimage and which is the image in question 1e?
- Is the preimage smaller or larger than the image?

1.7.2 Collaboration: Line Segments (continued)

- g. Draw a line that extends through the coordinate plane and passes through points M and S .
Shown on coordinate grid.
- h. Where do the two lines intersect?
The intersection is at the point $(-2.5, -3.5)$.
- i. What is a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{YZ}$?
Answers will vary. The transformation could be a translation up by 5. The transformation could be a reflection over the line $y = 3.5$. The transformation could be a 180° rotation at the point $(-1, 3.5)$.
- j. Ask two classmates the transformation they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Transformation	My Summary of Their Reason	Initials

2. Day'Ron and Zeb were discussing how to transform $\overline{TS}$ to create $\overline{YZ}$.

Day'Ron	Zeb
The only two single transformations are a translation of up 5, or a reflection across the line $y = 3.5$.	There are three single transformations. A translation of up 5, a reflection across the line $y = 3.5$, and a rotation of 180° .



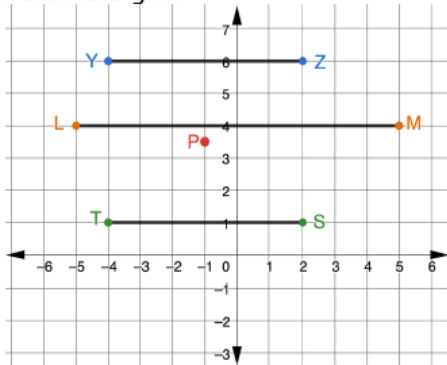
To provide organization to question 1j, consider having students “pair up” by finding “Soulmates.” Students will pair up with another student that has on shoes that match theirs. What defines a match is the student's own choice. It could be size, color, style, or anything the pair agrees on. After recording their “Soulmate’s” reasoning, students repeat this for finding a second “Soulmate.”



Students may benefit from having patty paper to explore the transformations that are discussed by Day'Ron and Zeb.

1.7.2 Collaboration: Line Segments (continued)

Day'Ron disagreed with Zeb. To prove his point, he used a graphing utility to show $\overline{TS}$, $\overline{YZ}$, $\overline{LM}$ and point P on the coordinate grid.



Day'Ron said, "It is not possible to rotate $\overline{TS}$ 180° so that point T aligns with point Y and point S aligns with point Z ."

Zeb replied, "Yes, but that does not matter because the letters do not have to match exactly. If you rotate $\overline{TS}$ 180° about point P it will coincide with $\overline{YZ}$."

Discuss Day'Ron and Zeb's justifications with your partner. Summarize your discussion.

Answers will vary. The resulting line segment if you rotate $\overline{TS}$ 180° about point P will coincide with $\overline{YZ}$, so it is possible for a 180° rotation about point P to be the transformation used.

3. Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed congruence?

Transformations to or from $\overline{TS}$ to $\overline{YZ}$ or $\overline{YZ}$ to $\overline{TS}$ are congruent.

4. Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed similarity?

Transformations involving $\overline{LM}$ are going to be similar.



To frontload for the next activity, ask students to begin discussing the differences between congruence and similarity.

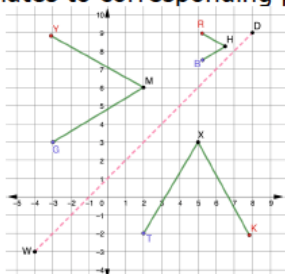
1.7.3 Guided Practice: Angles

Component Overview: Students are asked to describe a variety of transformations. Then, creating images of an angle, students determine that all transformations preserve angle measure.

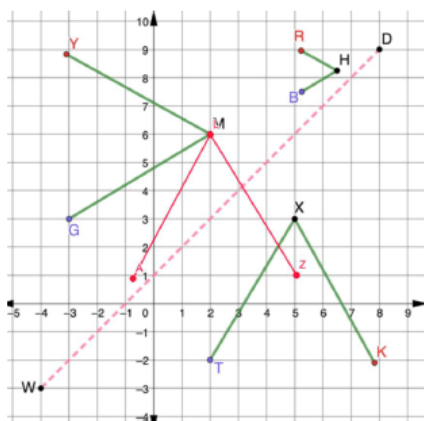


1.7.3 Guided Practice: Angles

$\overline{WD}$, $\angle YMG$, $\angle RHB$, and $\angle KXT$ are shown on the coordinate grid. Each color relates to corresponding points.



- Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle KXT$.
A reflection over $\overline{WD}$.
- Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle RHB$.
A dilation centered at point D.
- Use patty paper to create $\angle ALZ$ by rotating $\angle YMG$ 90° counterclockwise centered on point M.



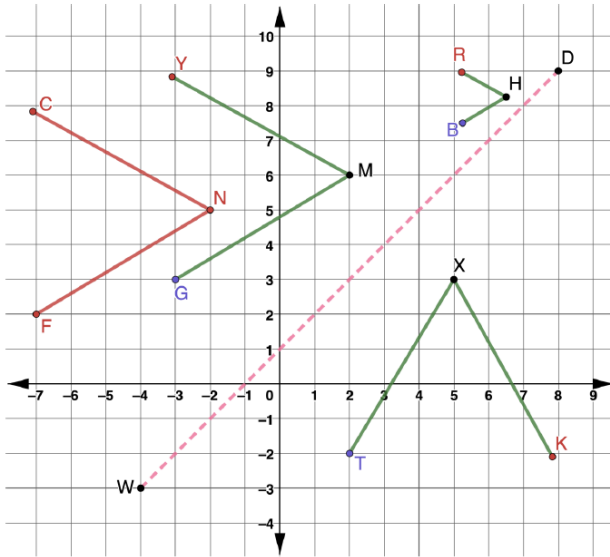
1.7.3 Guided Practice could be considered a guided discovery. Students are led to an understanding that all transformations preserve angle measure. Guiding this whole-group will help you monitor misconceptions and to ensure students understand that an angle's measure is not dependent on the length of the line segments.



Students move from transformations of line segments to transformations of angles. Allow students time to notice that the preimage and image of the angles are congruent. Determine if it is best for your students to pause after question 2 to ask students if the angles on the grid are congruent. Ensure that students do not “eyeball” it and instead use patty paper or a protractor to confirm congruence.

1.7.3 Guided Practice: Angles (continued)

4. Use patty paper to create $\angle CNF$ by translating $\angle YMG$ left 4 down 1.



Students should use the image copied on the patty paper to confirm that $\angle RHB$ and $\angle TXK$ both have the same angle measure as $\angle YMG$.

5. Which angles have the same measure as $\angle YMG$?
Angles $\angle KXT$, $\angle RHB$, $\angle ALZ$, and $\angle CNF$ all have the same angle measure as $\angle YMG$.
6. Which transformation(s) resulted in an angle whose measurement is the same as $\angle YMG$?
All transformations result in the same angle measure: rotation, reflection, translation, and dilation.

1.7.4 Bring It Together: Congruence and Similarity

Component Overview: Definitions of congruent and similarity.



1.7.4 Bring It Together: Congruence and Similarity

1. Complete the statements.

a. A rigid transformation results in a line segment that

- ☒ is
☐ is not

congruent to the original line segment.

b. A dilation results in a line segment that
congruent to the original line segment.

- ☐ is
☒ is not

c. A rigid transformation results in an angle that

- ☒ is
☐ is not

congruent to the original angle.

d. A dilation results in an angle that
congruent to the original angle.

- ☒ is
☐ is not



Two figures are **congruent** if they have exactly the same shape and size. Equivalently, two figures are congruent if one figure can be mapped to the other using a single rigid motion, or a sequence of transformations that includes only rigid motions.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.



Students will further explore a dilation of 1 and that it results in congruence in question 2. If students chose “is” in the second box, then consider asking them to explain.



To lead students through the process of generating a pictorial definition, consider asking students to underline the keywords of the definitions. This should reduce the number of words needed to define each word. After this, generate a picture of as many keywords as they can. Once they have these pictures, students should attempt to summarize these pictures, or use them to “create a scene” that would define the Key Term.

1.7.4 Bring It Together: Congruence and Similarity (continued)

Two congruent figures can also be considered similar with a scale factor of 1. Although this is known, it is not often required to state that two congruent figures are also similar.

2. Which statement in Question 1 would be false for a dilation with a scale of factor of 1?
Statement b would be false for a dilation with a scale factor of 1 would result in a congruent line segment.

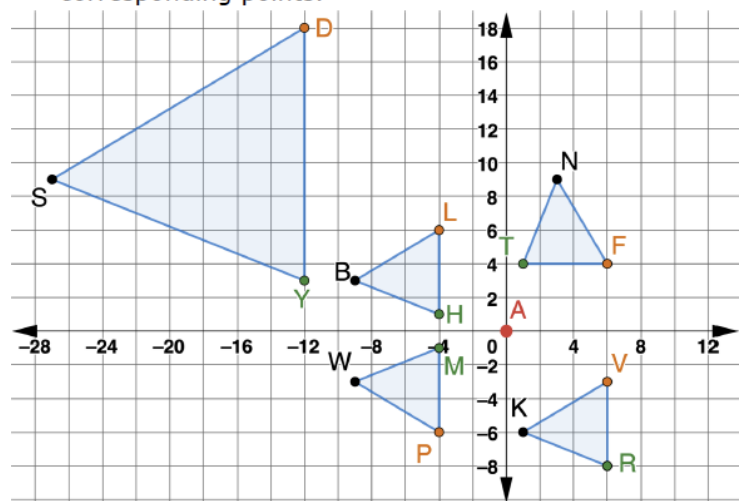
1.7.5 Guided Instruction: Congruence and Similarity Statements

Component Overview: Students look at the variety of figures and identify the transformation that could create each image, and if the images are congruent or similar to the preimage. Students then identify congruent or similar parts of the preimage and image.



1.7.5 Guided Instruction: Congruence and Similarity Statements

1. Five triangles are shown on the coordinate grid. $\triangle FNT$, $\triangle VKR$, $\triangle PWM$, and $\triangle DSY$ are created using a single transformation of $\triangle LBH$. Each color relates to corresponding points.



- a. Complete the table.

Triangle	Description of Transformation	Congruent or Similar
$\triangle FNT$	Rotation 90° Clockwise about point A	Congruent
$\triangle VKR$	Translation down 9 and to the right 10.	Congruent
$\triangle PWM$	Reflection across the x -axis.	Congruent
$\triangle DSY$	Dilation centered at point A with a scale factor of 3.	Similar



Take the time to show that the corresponding sides of the triangles are congruent or similar and that the corresponding angles are congruent. This lesson built up from line segments and angles to a triangle, so breaking the triangles to their parts lays the foundation for students to understand that the congruence or similarity of an image to its preimage can be considered by relating the corresponding line segments and angles.



Students may benefit from having patty paper or another manipulative that can be used to move $\triangle LBH$ to discover and describe the transformation.

1.7.5 Guided Instruction: Congruence and Similarity Statements (continued)

- b. Use the information about the transformations to complete each statement. *Answers may vary.*

$$\overline{LH} \cong \overline{MP}$$

$$\overline{TN} \cong \overline{HB}$$

$$\overline{YS} \sim \overline{BH}$$

$$\angle DYS \cong \angle LHB$$

$$\overline{MP} \sim \overline{YD}$$

$$\overline{FT} \cong \overline{LH}$$

$$\overline{PW} \cong \overline{BL}$$

$$\angle PMW \cong \angle LHB$$

$$\angle VKR \cong \angle LBH$$

- c. Discuss with your partner why there could be multiple answers to each statement. Summarize your discussion.
Answers will vary.



For a challenge, ask students to list all possible answers for each object in question 1b.

1.7.6 Wrap-Up: Cool Down

Component Overview: Students reflect on what they learned, what they did not know, and a question they still have.



1.7.6 Wrap-Up: Cool Down

Complete the statements about today's lesson.

1. Something I learned today ...

Answers will vary.

2. I didn't know ...

Answers will vary.



Answers to complete the statement "One question I still have ..." is where students may be asking for help, even if they do not say anything during class.

3. One question I still have ...

Answers will vary.